

Literature Review and Research Design

Module 2: Literature Review and Research Design

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INTRODUCTION

- The Review of Literature is a summary of all the review from various research literatures related to the current study carried out by a researcher
- It helps to discover what is already known about the research problem and what more has to be done
- According to Abdullah and Levin, the material gathered in literature review should be included as a part of the research data, since it influence the problem and research design it can be used to compare the result and findings for the current study



DEFINITION: LITERATURE REVIEW



Grove and Burns 2005

A literature review is an assessment of a body of research that addresses a research question.

An organized written presentation of what has been published on a topic by the scholars

Taylor 2011

An account of what has been published by accredited scholar and researchers.

CHARACTERISTICS OF A GOOD QUALITY OF LITERATURE REVIEW

- A good review of literature must be comprehensive
- It should include up to date references
- It should be systematic
- It should be reproducible
- It should be free from bias
- It should be well-written
- It should be in the form of some of its parts
- It should be clearly searched and selected
- Accurate references should be given in the review



FACTORS AFFECTING REVIEW OF LITERATURE

Researcher's Background

An experienced researcher finds it easy to prepare a review of literature than a beginner.

Complexity of the research project

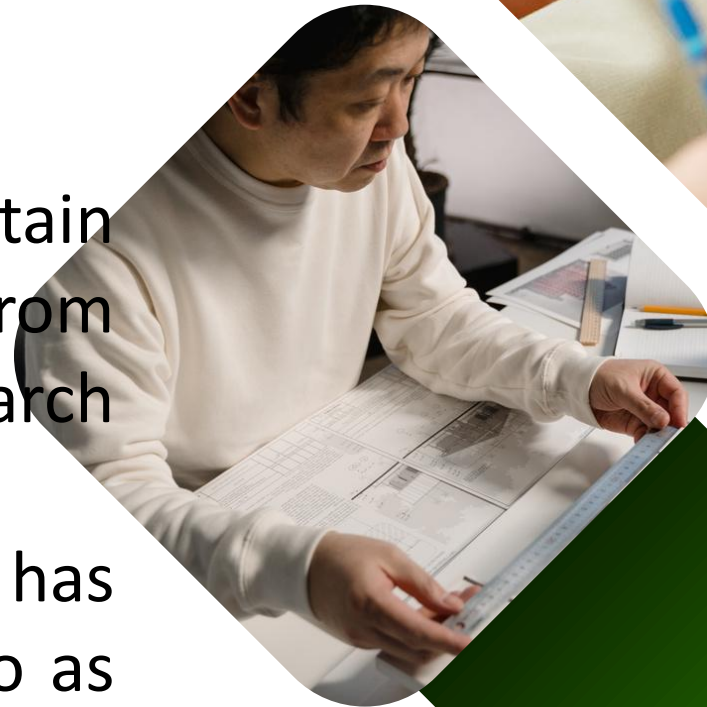
It is easy to collect review of literature for simple and easy research projects than complex ones.

Availability of resources

Availability of resources like computer, internet facility, online and offline journal subscription makes it easy to prepare a good review of literature.

PURPOSE OF REVIEW OF LITERATURE

- It helps to identify what is already known about a research problem.
- It helps to build on previous knowledge.
- Many published research studies contain recommendation for future research, from which we can get idea for a new research study.
- It makes researcher know what research has already been done in a particular area so as to avoid duplication.



PURPOSE OF REVIEW OF LITERATURE

- It is necessary to narrow the problem to be studied.
- It helps researcher to get acquainted with the relevant theory, strategies, tools and instruments for conducting the research.
- It helps to understand the different ways of conducting the research study.
- It helps to identify comparative data so helps in interpretation and discussion of previous study.



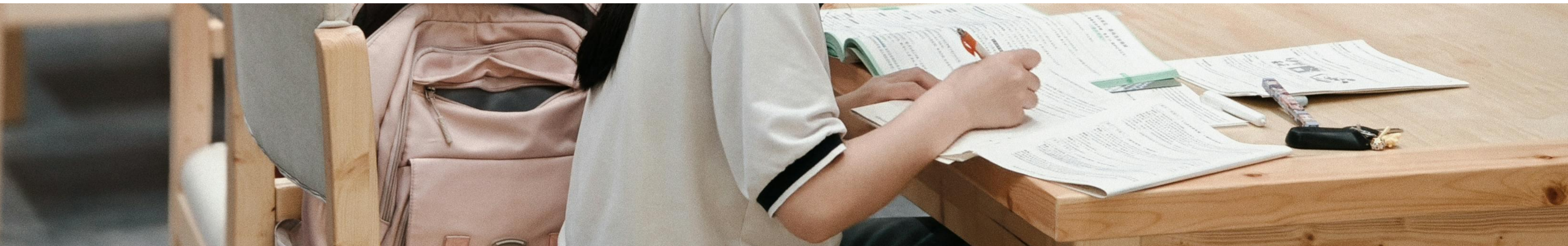
SOURCES OF REVIEW OF LITERATURE

PRIMARY SOURCE:

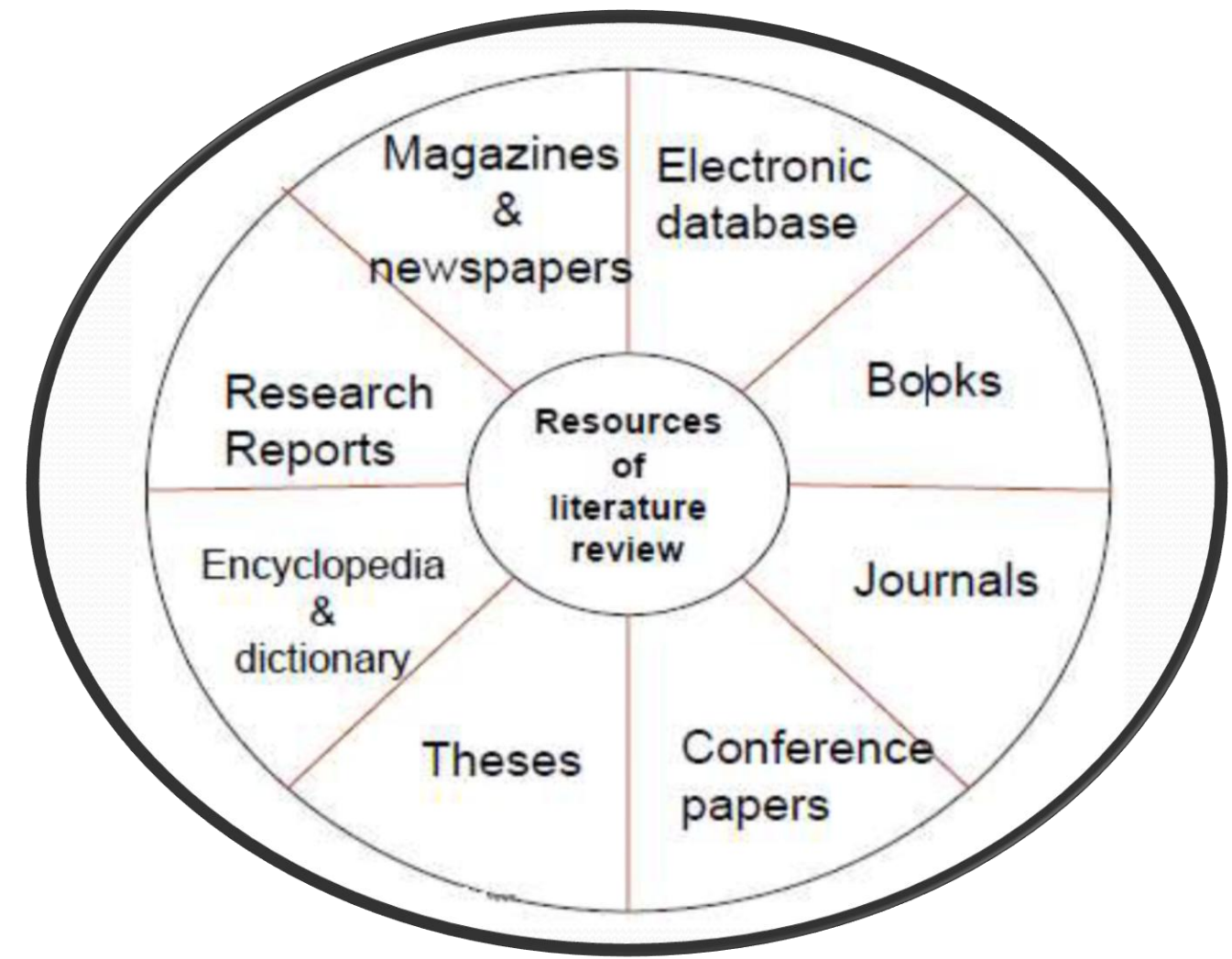
A primary Source is a literature resource of a study written by the original researcher. It is research publication written by the person or people who conducted the research.

SECONDARY SOURCE:

These are second hand information prepared or written by someone other than the original author.
Examples: Newspaper, Book Chapters, Television, Radio, Magazine, Wikipedia, Journals, etc.



RESOURCES OF LITERATURE REVIEW



CS SPECIFIC DIGITAL LIBRARIES

- ACM Digital Library
- Google Scholar
- IEEE Explore
- Science Direct
- Web of Science
- Taylor and Francis



THREE PASS READING

PASS 1 HIGH LEVEL OVERVIEW

What to read: Title, Abstract, and Introduction.

What to skim: Section headings, subheadings, and math formulas (just notice what is there).

What to finish with: Read the Conclusion section.

Check figures/tables: Look at key diagrams or main performance charts.

PASS 2 THE CORE ANALYSIS

Examine visuals closely: Look at graphs, charts, and diagrams. Pay attention to axis labels, error bars, and baseline comparisons.

Mark unread literature: Note unfamiliar terms, algorithms, or references you need to look up later.

Critical evaluation: Ask yourself, "Did the authors prove what they claimed? Are the baselines fair?"

PASS 3 VIRTUAL RE-IMPLEMENTATION

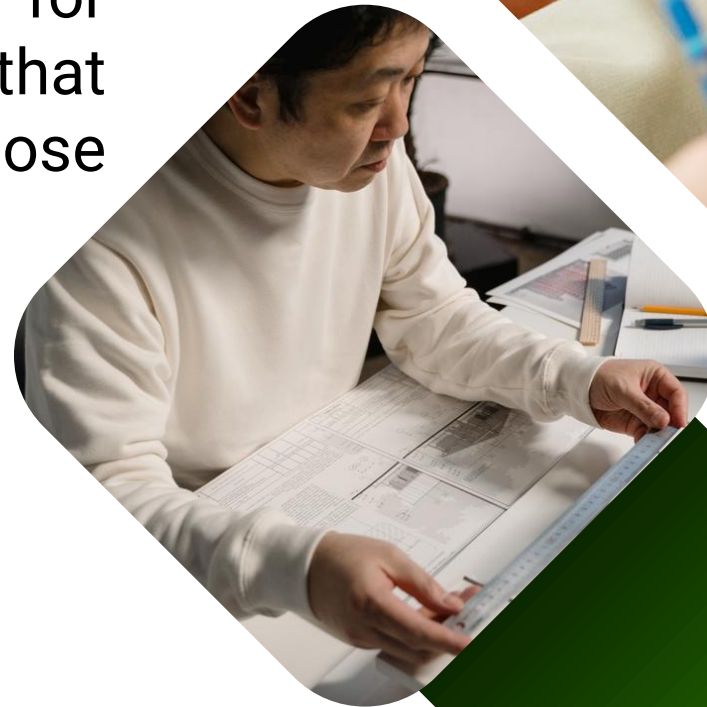
Challenge every assumption: Try to predict what the authors did at each step before reading their solution.

Verify calculations: Check equations, proofs, algorithmic steps, and experimental setups line by line.

Identify hidden gaps: Spot unstated assumptions, edge cases the authors ignored, or potential failure modes.

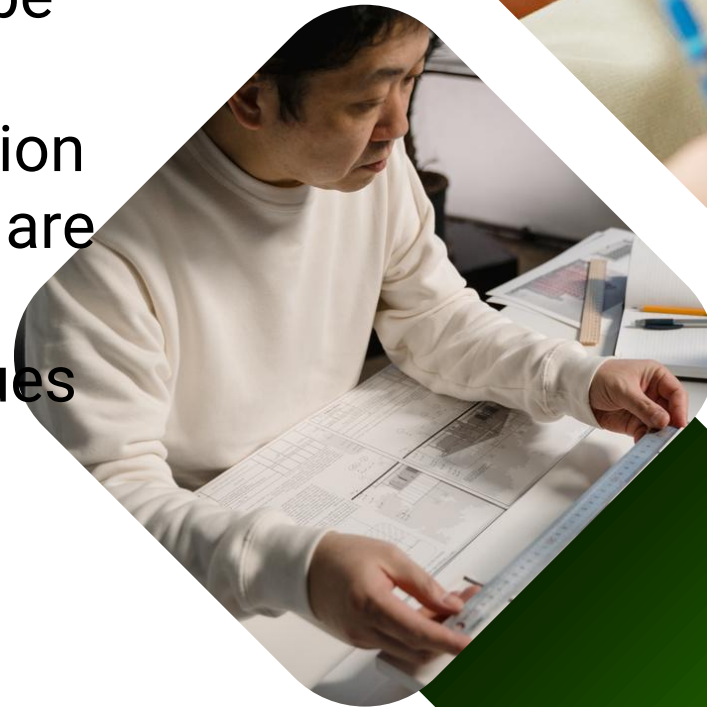
RESEARCH DESIGN

Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitute a research design. "A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure.



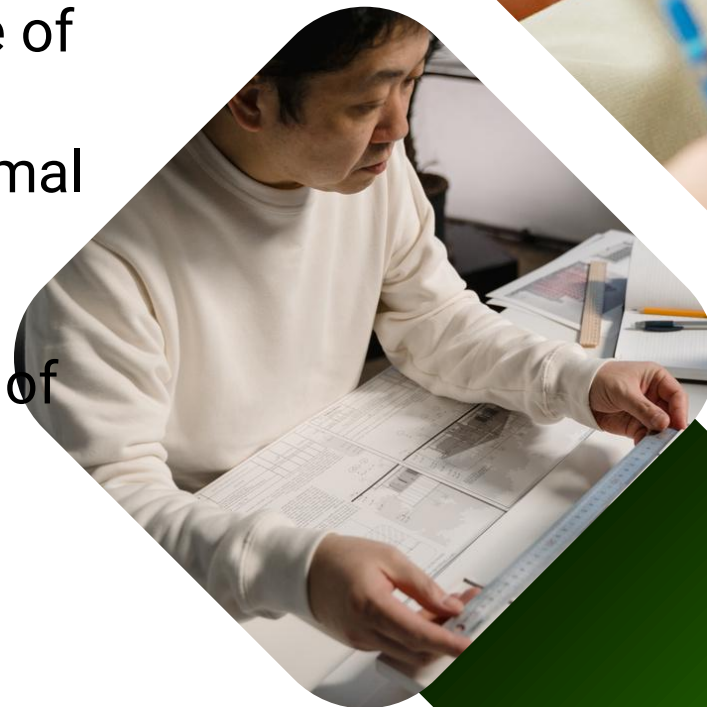
RESEARCH DESIGN

1. Sampling design: which deals with the methods of selecting items to be observed for the study.
2. Observational Design: which relate to the condition under which the observation are to be created
3. Statistical design: Which concerned the question of the how the information and data gathered are to be analyzed
4. Operational Design: which deals with techniques by which procedure satisfied in sampling



IMPORTANT CONCEPT RELATING TO RESEARCH DESIGN

- A concept which can take on different quantitative values is called a variable. As such the concepts like weight, height, income are all examples of variables. Qualitative phenomena (or the attributes) are also quantified on the basis of the presence or absence of the concerning attribute(s). Phenomena which can take on quantitatively different values even in decimal points are called 'continuous variables'.
- If one variable depends upon or is a consequence of the other variable, it is termed as a dependent variable, and the variable that is antecedent to the dependent variable is termed as an independent variable.



Independent Variable: Learning rate hyperparameter (e.g., 0.1, 0.01, 0.001, 0.0001).

Dependent Variable: Model classification accuracy (%) and training time to reach convergence (number of epochs).

IMPORTANT CONCEPT RELATING TO RESEARCH DESIGN

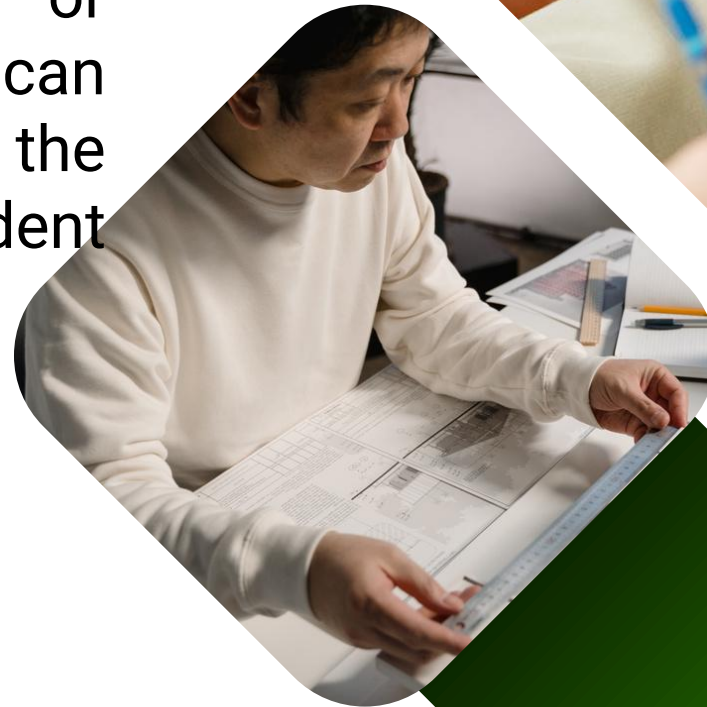
Extraneous Variable: An **extraneous variable** is any uncontrolled variable that is *not* the focus of the study, but has the potential to influence the outcome (dependent variable). If not controlled, it adds random noise or variance to the data.

Environment / Testing Conditions: Testing one group of students in a quiet, air-conditioned room and another group in a noisy, hot room during an intelligence test. Room temperature and noise level are extraneous variables.



IMPORTANT CONCEPT RELATING TO RESEARCH DESIGN

Control: Control in research design refers to the strategies researchers use to minimize the influence of extraneous and confounding variables. By holding extraneous factors constant or balancing them across conditions, researchers can isolate the true relationship between the independent variable (cause) and the dependent variable (effect).



IMPORTANT CONCEPT RELATING TO RESEARCH DESIGN

Confounded relationship: A **confounded relationship** occurs when an extraneous variable systematically varies *alongside* the independent variable. As a result, you cannot determine whether the independent variable or the confounding variable caused the effect on the dependent variable.

Educational Tech Intervention Study:

Independent Variable (X): A new interactive math software vs. traditional paper worksheets.

Dependent Variable (Y): Final exam scores.

Confound: The software group is assigned to an 8:00 AM class taught by Teacher A, while the worksheet group is assigned to a 2:00 PM class taught by Teacher B.



IMPORTANT CONCEPT RELATING TO RESEARCH DESIGN

Research hypothesis: When a prediction or a hypothesised relationship is to be tested by scientific methods, it is termed as research hypothesis. The research hypothesis is a predictive statement that relates an independent variable to a dependent variable.

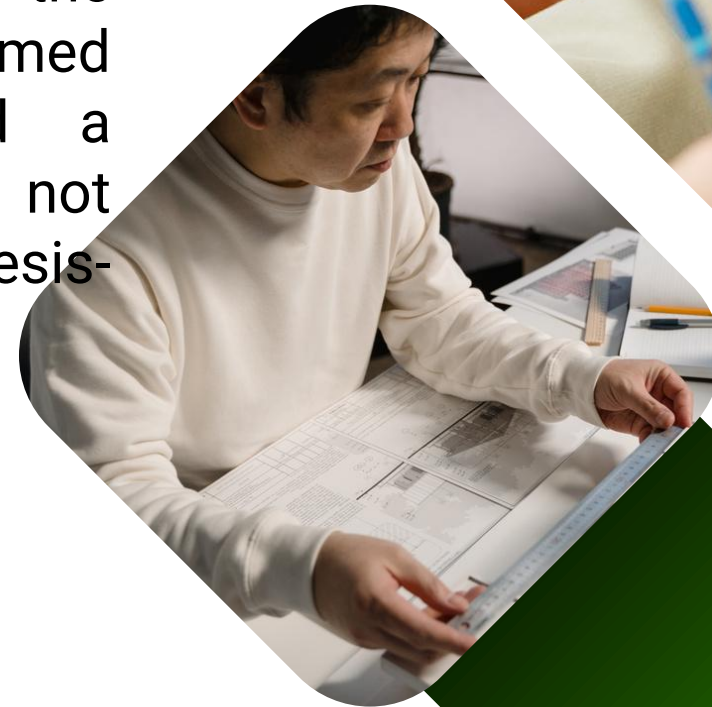
Example: Testing a new cache-replacement algorithm against the industry standard (LRU).

Directional Hypothesis (H_1): *Under heavy write workloads, a dynamic frequency-based cache replacement algorithm will achieve a lower cache-miss rate than the Least Recently Used (LRU) algorithm.*



IMPORTANT CONCEPT RELATING TO RESEARCH DESIGN

Experimental and non-experimental hypothesis-testing research: When the purpose of research is to test a research hypothesis, it is termed as hypothesis-testing research. It can be of the experimental design or of the non-experimental design. Research in which the independent variable is manipulated is termed 'experimental hypothesis-testing research' and a research in which an independent variable is not manipulated is called 'non-experimental hypothesis-testing research'.



DIFFERENT RESEARCH DESIGNS

EXPLORATORY RESEARCH STUDIES

Exploratory research studies are also termed as formulative research studies. The main purpose of such studies is that of formulating a problem for more precise investigation or of developing the working hypotheses from an operational point of view.

The major emphasis in such studies is on the discovery of ideas and insights. As such the research design appropriate for such studies must be flexible enough to provide opportunity for considering different aspects of a problem under study.

Investigating Why Freshmen Drop Out of Online Coding Courses



DIFFERENT RESEARCH DESIGNS

DESCRIPTIVE AND DIAGNOSTIC RESEARCH STUDIES

Descriptive research studies are those studies which are concerned with describing the characteristics of a particular individual, or of a group, whereas diagnostic research studies determine the frequency with which something occurs or its association with something else. The studies concerning whether certain variables are associated are examples of diagnostic research studies.

Survey of Tech Stacks Used in BCA Final Year Projects



CONCLUSION

SUMMARY

There are several research designs and the researcher must decide in advance of collection and analysis of data as to which design would prove to be more appropriate for his research project.



THANK YOU

WE APPRECIATE YOUR TIME AND
ATTENTION TO OUR RESEARCH STUDY.

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